

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and maintained **data pipelines** using **Python**, **SQL**, and **AWS** (EC2, S3, Lambda) to ingest, transform, and validate tens of millions of financial transaction records daily; reduced data ingestion latency by **35%** through optimized ETL workflows and CloudWatch monitoring.
- Built scalable **data inspection tools and services** using **Python** and **REST APIs** to detect data quality issues, biases, and error patterns across production systems; deployed to **AWS** cloud infrastructure supporting **99.9% SLA** uptime for mission-critical payments operations.
- Developed interactive **data visualization dashboards** using **Power BI** and **Tableau** to surface model quality metrics, annotation coverage, and data anomalies to cross-functional teams of data scientists, machine learning researchers, and domain taxonomists; enabled **40% faster** root-cause analysis of data pipeline failures.
- Applied **software engineering rigor** and **CI/CD best practices** via **Jenkins** automation pipelines and **Git** version control to deploy **database designs** and robust, scalable microservices to production; implemented automated testing and code review workflows reducing deployment defects by **28%**.
- Integrated **3rd-party systems and APIs** across **cloud environments** (AWS) to connect annotation platforms, crowdsourcing tools, and ontology services; built **API/SDK interactions** in **Python** to orchestrate data flows between tools used by data science teams, supporting real-time data validation at scale.
- Collaborated with **Agile** cross-functional teams (data engineers, linguists, knowledge engineers, ML researchers) across multiple time zones using **Jira** and **Confluence** for sprint planning and documentation; delivered **12+ data infrastructure features** supporting advertising inventory verification and brand safety ontology expansion.

SKILLS

Data Engineering & Pipelines – data pipeline design, data pipeline maintenance, ETL, SQL (PostgreSQL, MySQL, Oracle), Python, Spark, Databricks.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), cloud environments, CI/CD automation, Jenkins, Github, Git.

Data Quality & Inspection – data quality tools, annotated data inspection, bias detection, error pattern investigation, data validation, database design.

Data Visualization & Analytics – Power BI, Tableau, dashboards, DAX, Power Query, data modelling.

Software Engineering – REST APIs, API/SDK integration, 3rd-party system integration, Python scripting, microservices, scalable software design.

Collaboration & Agile – Agile, cross-functional teamwork, Jira, Confluence, technical documentation, communication.

Big Data & Frameworks – Spark, Big Data frameworks, high-throughput systems, real-time data processing.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Real-Time Annotation Quality Data Pipeline & Inspection Platform (HCL Technologies)

- Architected a **large-scale data pipeline** using **Python, Spark on Databricks**, and **AWS** (S3, Lambda, CloudWatch) to process **500M+ daily annotation records** from crowdsourcing platforms; implemented **SQL-driven** schema validation and data lineage tracking reducing data quality incidents by **45%**.
- Developed **scalable inspection tools and services** in **Python** to automatically detect biases, labeling errors, and anomalies in annotated datasets; exposed results via **REST APIs** enabling downstream data science tools and crowdsourcing platforms to consume quality signals in real-time.
- Built interactive **Power BI dashboards** and **Tableau visualizations** displaying model quality metrics, annotation coverage gaps, and error-pattern trends; enabled taxonomists and ML researchers to identify problematic ontology categories within **48 hours** vs. **2+ weeks** prior.
- Implemented **CI/CD pipelines** in **Jenkins** with **Github** integration to automate data pipeline deployments, unit testing, and schema validation; reduced time-to-production for new data quality checks from **5 days to 12 hours**.
- Collaborated with knowledge engineers, linguists, and data scientists to gather requirements for data inspection features; documented pipeline architecture and API contracts in **Confluence** enabling smooth handoffs and reducing onboarding time for new team members by **60%**.

SWIFT Messaging Operations Dashboard & Payments Data Quality Platform (HCL Technologies)

- Designed and maintained **data pipelines** ingesting **ISO 20022 message flows** and **SWIFT MT/MX formats** from real-time payments operations; built **SQL-backed** validation layers in **PostgreSQL** and **Oracle** to detect transaction anomalies and regulatory compliance breaches across **SEPA instant payments** and **cross-border payment** corridors.
- Developed **Power BI dashboards** visualizing transaction flows, SLA metrics, and payment-status insights for banking operations teams and compliance auditors; surfaced **GDPR data retention** compliance indicators and tokenization events reducing audit discovery time by **50%**.
- Integrated **3rd-party payment systems and APIs** (Open Banking/PSD2 endpoints) with **REST APIs** and **AWS** services to orchestrate data flows between core payments processors, reconciliation engines, and business intelligence platforms supporting **99.95% message delivery accuracy**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)